Efficiency

Exam code: 0625 & 0972

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Help me understand

Efficiency of energy transfer

- The **efficiency** of a system is a measure of the amount of **useful** and **wasted energy** in an energy transfer
- Efficiency is defined as:

The ratio of the useful power or energy output from a system to its total power or energy input

- If a system has **high** efficiency, this means most of the energy transferred is **useful**
- If a system has **low** efficiency, this means most of the energy transferred is **wasted**

Calculating efficiency

Extended tier only

- Efficiency is represented as a percentage and can be calculated using two equations
- Efficiency in terms of **energy**:

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \ (\times 100\%)$$

- Efficiency in terms of **power**:

$$\text{efficiency} = \frac{\text{useful power output}}{\text{total power input}} \ (\times 100\%)$$

Stuck on something?

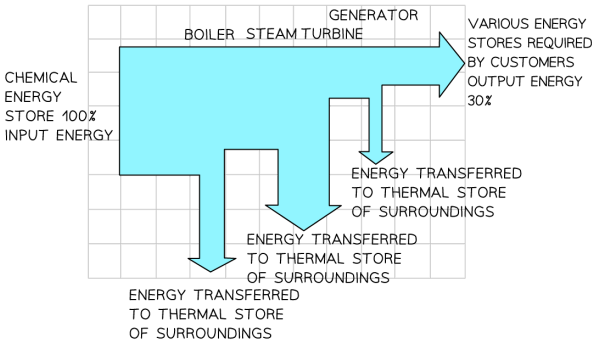
Ask questions and get help with exactly what you're revising.

Got it

- The turbine turns a generator
- The generator produces electricity
  - At each stage of this process, energy is dissipated to the surroundings

- The overall efficiency of a typical thermal power station is approximately 30%
  - This means that 70% of the energy transferred from the power station to the National Grid is **wasted energy**

Sankey diagram of electricity production



Sankey diagram showing the energy transfers involved in generating electricity in a gas-fired power station

Worked Example

An electric motor lifts a 7.2 kg load through a height of 5.0 m in 3 seconds. The efficiency of the motor is 35%.

Calculate the power supplied to the motor.

Answer:

Step 1: List the known quantities

- Mass of load, m = 7.2 kg
- Change in height, h = 5.0 m
- Acceleration of free fall, g = 9.8 m/s^2
- Time, t = 3 s
- Efficiency = 35% or 0.35

Step 2: Calculate the power output of the motor

- The power supplied by the motor is the work done, or energy transferred, to lift the load

$$\text{power output} = \frac{\Delta E}{t}$$

- ΔE is equal to the change in gravitational potential energy as the load is lifted

$$\Delta E_P = mg\Delta h$$

$$\text{power output} = \frac{352.8}{3} = 117.6 \text{ W}$$

Step 3: Write down the efficiency equation in terms of power

$$\text{efficiency} = \frac{\text{useful power output}}{\text{total power input}}$$

Step 4: Rearrange to make power input the subject

$$\text{power input} = \frac{\text{power output}}{\text{efficiency}}$$

Step 5: Substitute the values into the power input equation

$$\text{power input} = \frac{117.6}{0.35}$$

$$\text{power input} = 336 \text{ W}$$

Examiner Tips and Tricks

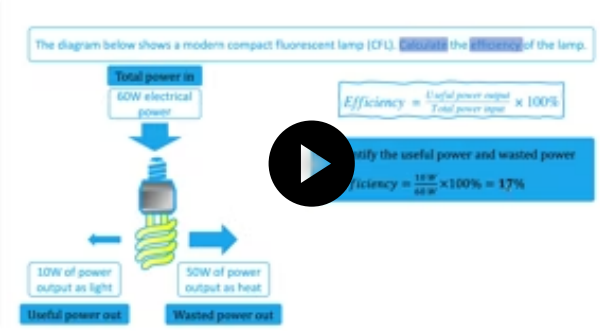
Efficiency can be given in a ratio (between 0 and 1) or percentage format (between 0 and 100 %)

If the question asks for efficiency as a ratio, give your answer as a fraction or decimal.

If the answer is required as a percentage, remember to multiply the ratio by 100 to convert it:

- if the ratio = 0.25, percentage = 0.25 × 100 = 25 %

Remember that efficiency has **no units**



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